

Air Convection

Student sheet

Name: _____

Date: _____

1. What I see on the screen

The screen shows **Temperature** (*in °C*) and **Humidity** (*in %*), and below them their chart over time. **Watch the shape of the curve, not just the number.** You start with **Start**; you move on with the green button at the bottom right, you go back with **Back**, and the last green button reads **Finish**.

The sensor is slow: 60–120 s for T , up to 3 min for humidity. It stays 10–15 cm above the source, never touching it.

2. What I do on my own

Source (circle one): container of warm water / lamp / radiator **Heating on?** ☐ YES ☐ NO
First I predict. I do not change my prediction after I measure.

1. Near the ceiling the air is ☐ warmer ☐ cooler ☐ the same as near the floor, by _____ °C.
2. Above the source, the humidity will ☐ rise ☐ fall ☐ stay the same

because _____

3. I draw the arrows. On the left, *before* any measurement: how do you think the air moves around the source? On the right, at the end, what you saw. **The loop must close.**

What I predict (before)

What I saw (after)

Now I measure. I wait at every position. Shape of the curve: rises fast, rises slowly, has stopped, falls.

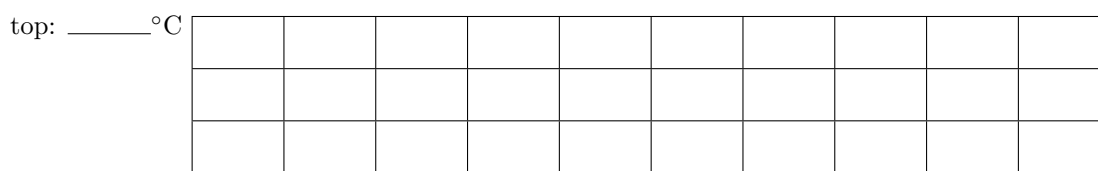
Position	T (°C)	RH (%)	Shape of the curve
Reference, on the desk			flat
15 cm above the source			
Back on the desk			

Difference (above the source – reference) = _____ – _____ = _____ °C

The vertical profile. At least 90 s at every height; heights are *measured*, not guessed.

Height	T (°C)	Compared to the desk
0 cm (floor)		☐ warmer ☐ cooler
50 cm		☐ warmer ☐ cooler
100 cm		☐ warmer ☐ cooler
150 cm		☐ warmer ☐ cooler
200 cm		☐ warmer ☐ cooler

The graph. The temperature axis does not start at zero.



bottom: _____ °C Height above the floor, from 0 to 200 cm — ten divisions of 20 cm each

The gradient. Top _____ – floor _____ = _____ °C, over _____ m, so _____ °C/m; in a sports hall 4 m high it would be _____ °C, but my prediction could be wrong because _____

3. The conclusion I write

I predicted that _____

The data show that _____

Warm air rises because _____

Exit ticket

One thing that surprised you today, and one thing you would do differently if you repeated the measurement.
